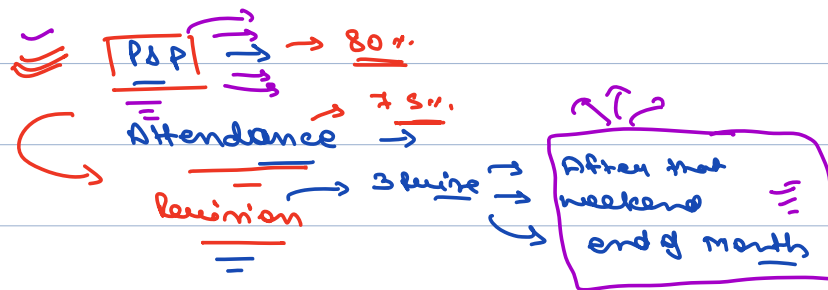


I wanted the reward and not the struggle. I wanted the result and not the process. I was in love with not the fight but only the victory.

Mark Manson



Intermediate module

- Introduction to Problem Solving
- Time Complexity
- Introduction to Arrays
- Prefix Sum
- Carry Forward
- Subarrays
- 2D Matrices
- Sorting Basics
- Hashing Basics
- Strings Basics
- Bit Manipulation Basics
- Interview Problems
- Contest [covers Full Intermediate DSA]

Today's Content

1. Count the Factors
2. Optimisation for counting the Factors
3. Check if a number is Prime
4. Sum of N Natural Numbers
5. Definition of AP & GP
6. How to find the number of a times a piece of code runs, i.e, number of Iterations.
7. How to compare two Algorithms.

Ques) what is a factor?

i is a factor of $n \rightarrow ?$

$$\underline{n \% i == 0}$$

2 is factor of 10

$\rightarrow$ yes.

Ques) Given n , count factors of n . ($n > 0$).

Number of factors of the number 24. $\rightarrow$ 1, 2, 3, 4, 6, 8, 12, 24.
8 factors.

Number of factors of the number 10. $\rightarrow$ 1, 2, 5, 10
4 factors.

n $\rightarrow$ smallest factor $\rightarrow$ 1.
 n $\rightarrow$ largest factor $\rightarrow$ n
 $\rightarrow$ factors of $n \rightarrow$ $[1, n]$.

function count_factors(n) {

$count = 0$;

 for $i \rightarrow 1$ to n :

 if ($n \% i == 0$) {

$count++$

 return $count$;

10^8 iterations in 1 sec.

N	iteration	Time
10^8	10^8	1 sec
10^9	10^9	10 sec
10^{18}	10^{18}	10^{10} sec ≈ 317 yrs.

$$10^8 \text{ iter} \rightarrow 1 \text{ sec}$$

$$1 \text{ iter} = \frac{1}{10^8} \text{ sec}$$

$$10^9 \text{ iter} = \frac{1}{10^8} \times 10^9 \text{ sec}$$

Optimization :-

$$J = \frac{N}{i}$$

$i * J = N$ $\{i \text{ \& } J \text{ are factors of } N\}$

$6 \times 3 = 18$ $\{6 \text{ \& } 3 \text{ are factors of } 18\}$

$7 \times 8 = 56$ $\{7 \text{ \& } 8 \text{ are factors of } 56\}$

i	$\frac{N}{i}$	<u>$N=24$</u>	
1	24		
2	12		
3	8		
4	6		
6	4		
8	3		
12	2		
24	1		

$\sqrt{N}$

$$i^2 \leq N$$

$$i \leq \sqrt{N}$$

i	$\frac{N}{i}$	<u>$N=100$</u>	
1	100		
2	50		
4	25		
5	20		
10	10		
20	5		
25	4		
50	2		
100	1		

N = 24

1 to 4

1

count
+2

2

+2

3

+2

4

+2

N = 40

1 to 6

1

Count

2

+2

(1 2 40)

2

+2

(2 2 20)

4

+2

(4 2 10)

5

+2

(5 2 8)

function countFactors(n) {

count = 0;

for i: 1 to $\sqrt{n}$ ✓ 1 to 10

if (n % i == 0) {

count += 2

X

return count;

$$N = 100$$

1	+2	(1 & 100)
2	+2	(2 & 50)
4	+2	(4 & 25)
5	+2	(5 & 20)
10	+2	(10 & 10)

function count factors (N) {

count = 0;

$i \leq N$

for (i = 1; i <= $\sqrt{N}$; i++) {

if (N % i == 0) {

if (i == N/i) { count++; }

else { count += 2; }

return count;

N
10¹⁸

iterations
10⁹

time
10sec

$$\sqrt{x} \rightarrow x^{1/2}$$

$$\sqrt{10^{18}} \rightarrow 10^{18 \times \frac{1}{2}} \Rightarrow \underline{10^9}$$

Break 10:02 pm - 10:17 pm

Ques) Given n , check if it's prime or not.

How many prime numbers are there?

10, 11, 23, 2, 25, 27, 31

no. of factors → 4, 2, 2, 2, 0, 4, 2

Prime → no. which has 2 factors.

1 → 1 x

2 → 1 2 ✓

function isPrime(n) {

if (countFactors(n) == 2) {

return True

}

else

return false

}

}

$$S = 1 + 2 + 3 + \dots + 99 + 100$$

$$+ S = 100 + 99 + 98 + \dots + 2 + 1$$

$$2S = 101 + 101 + 101 + \dots + 101 + 101$$

100 times

$$2S = 101 \times 100$$

$$S = \frac{101 \times 100}{2} =$$

Generalize

Sum of 1st n natural no.

$$S = 1 + 2 + 3 + \dots + (n-1) + n$$

$$S = n + (n-1) + (n-2) + \dots + 2 + 1$$

$$2S = (n+1) + (n+1) + (n+1) + \dots + (n+1) + (n+1)$$

n times

$$2S = n \times (n+1)$$

$$S = \frac{n \times (n+1)}{2}$$

Some basic math properties :-

$[a, b]$

(a, b)

$[2, 4]$

1. $[a, b]$ - This type of range means that a and b are both inclusive.
2. (a, b) - This type of range means that a and b are both excluded.

How many numbers are there in the range $[3, 10]$?

$$\begin{aligned} & \rightarrow 10 - 3 + 1 \Rightarrow 8 \\ & \downarrow \\ & 3, 4, 5, 6, 7, 8, 9, 10 \rightarrow 8 \text{ no's.} \end{aligned}$$

$$[a, b] \Rightarrow b - a + 1$$

$\rightarrow$ Total numbers from a to b.

Iteration $\rightarrow$ The no. of times a loop is run.

Quiz :-

How many times will the below loop run ?

```
for(i=1; i<=N; i++)  
{  
    if(i == N) break;  
}
```

$[1 \text{ to } N] \rightarrow N - 1 + 1 \Rightarrow N \text{ iterations.}$

Ques)

How many iterations will be there in this loop ?

```
for(int i = 0; i <= 100; i++){  
    s = s + i + i^2;  
}
```

$i \rightarrow 0 \text{ to } 100 \rightarrow 100 - 0 + 1 \Rightarrow 101.$

Quiz

How many iterations will be there in this loop?

```
func(){  
    for(int i = 1; i <= N; i++){  
        if(i % 2 == 0){  
            print(i);  
        }  
    }  
    for(int j = 1; j <= M; j++){  
        if(j % 2 == 0){  
            print(j);  
        }  
    }  
}
```

$1 \text{ to } N \Rightarrow N - 1 + 1 \Rightarrow N$

$1 \text{ to } M \Rightarrow M - 1 + 1 \Rightarrow M$

$N + M$ iterations.

Geometric Progression (G.P.)

5, 10, 20, 40, 80 = $5 \frac{(2^5 - 1)}{2 - 1}$

$\frac{10}{5} = \frac{20}{10} = \frac{40}{20} = \frac{80}{40} \Rightarrow 2 \rightarrow$ Common ratio.

// Generic Notation:-

$a, ar, ar^2, \dots$

$\hookrightarrow$ n terms of G.P.

Sum $\Rightarrow \frac{a(r^n - 1)}{r - 1} \quad (r \neq 1)$

$a, a, a, \dots$

$\underbrace{\hspace{1.5cm}}_n$

$\text{sum} = \underline{a \cdot n}$



σ_{n-1}

Arman

Vikay

Sort an array.

$\text{arr}[5] = \{3, 2, 6, 8, 13\} \Rightarrow \{1, 2, 3, 6, 8, 13\}$

Arman

15 sec



(Windows XP)



MacBook m2



7 sec



C++



(Top of Hot Volcano)



m+ Everest

(3 sec)

Vikay

10 sec



(MacBook m2)



10 sec



Python



C++

5 sec

Conclusion

We can't evaluate algorithm's performance using **execution time** as it depends on a lot of factors like operating system, place of execution, language, etc.

Question

How can we compare two algorithms?

Which measure doesn't depend on any factor?

no. of iterations.

- The number of iterations of an algorithm remains the same irrespective of Operating System, place of execution, language, etc.

next class :-

↳ Big OH.

↳ logarithms.

↳ Space Complexity.

↳ The Importance of constraints.

whatsapp.

8383803711

2, 4, 6, 8, 10, ...

$\underbrace{\quad}_2 \underbrace{\quad}_2 \underbrace{\quad}_2 \underbrace{\quad}_2$

$$S = \sum [2a + (n-1)d]$$

$$a, ar, ar^2 + \dots ar^{n-1}$$

$$S = a + \cancel{ar} + \cancel{ar^2} + \dots + \cancel{ar^{n-1}}$$

$$- \quad rS = \quad \cancel{ar} + \cancel{ar^2} + \cancel{ar^3} + \dots + \cancel{ar^{n-1}} + ar^n$$

$$S - rS = a - ar^n$$

$$S(1-r) = a(1-r^n)$$

$$S = \frac{a(1-r^n)}{1-r} \Rightarrow S = \frac{a(r^n-1)}{r-1}$$